Structured Probabilistic Models for Hierarchical Clustering of Individual Stocks or Market Indices

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ABSTRACT (Extended Abstract)

Financial markets behave as complex adaptive systems in which asset returns are driven not only by firm-level information but also by shared macroeconomic conditions, sectoral factors, and collective investor behavior. Because of these interdependencies, analyzing assets individually often fails to capture the underlying market structure. This study proposes a structured probabilistic framework combining correlation networks, probabilistic filtering, and hierarchical clustering to identify latent organization among financial assets.

Historical price data are transformed into logarithmic returns to allow consistent comparison across assets. A correlation matrix is then computed to measure statistical dependence and represented as a weighted financial network in which nodes correspond to assets and edges capture co-movement strength. Since empirical correlation matrices contain significant noise, probabilistic filtering methods such as Minimum Spanning Trees (MST) and bootstrap reliability analysis are applied to retain only statistically stable relationships (Bonanno et al., 2003; Tumminello et al., 2007). Correlations are transformed into metric distances and analyzed using agglomerative hierarchical clustering to reveal nested groupings without requiring predefined cluster counts (Mantegna, 1999).

This draft incorporates revisions from peer feedback and proposal modifications, including clearer explanation of bootstrap validation, improved methodological flow, and stronger emphasis on practical financial interpretation. Preliminary findings indicate cluster structures broadly align with economic sectors and that higher market volatility produces tighter clustering, suggesting reduced diversification effectiveness. The study contributes to financial data science by integrating probabilistic modeling, network theory, and machine learning into a unified and interpretable framework for analyzing market structure and systemic risk.

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1. Introduction

Financial markets are characterized by strong interdependence among assets due to shared exposure to economic conditions and investor sentiment. Traditional analysis methods frequently evaluate securities independently, yet research shows that financial markets exhibit collective behavior that emerges through correlation structures (Mantegna, 1999). Representing markets as networks provides a way to model these interactions and better understand systemic behavior.

Network-based methods allow relationships between assets to be visualized and quantified, making it possible to observe how shocks propagate across the system (Billio et al., 2012). Hierarchical clustering further reveals multi-level organization such as sector-level groupings and broader market regimes (Tumminello et al., 2010). This project explores whether structured probabilistic modeling improves stability and interpretability of clustering outcomes.

2. Background

Early studies in econophysics demonstrated that correlations among asset returns can be transformed into distance metrics that reveal hierarchical organization in markets (Mantegna, 1999). This approach showed that groups of stocks naturally cluster based on economic similarity without requiring predefined sector labels.

However, correlation matrices derived from financial data are inherently noisy due to sampling limitations and evolving market conditions. To address this issue, filtering techniques such as Minimum Spanning Trees were introduced to retain the most informative relationships while reducing complexity (Bonanno et al., 2003). Bootstrap methods were later proposed to

evaluate statistical reliability of network edges, ensuring that observed structures are stable rather than random artifacts (Tumminello et al., 2007).

Together, these developments provide a strong foundation for combining probabilistic filtering and hierarchical clustering in financial research.

3. Problem Statement

Financial markets are highly interconnected systems in which asset prices are influenced by shared economic conditions, sector trends, and investor behavior. Although correlation analysis and clustering techniques are commonly used to study these relationships, traditional approaches often produce unstable or difficult-to-interpret results. One key reason is that financial correlation matrices are typically noisy due to limited sample sizes, changing market environments, and short-term fluctuations. As a result, clustering methods applied directly to raw correlations may capture random noise rather than meaningful market structure.

Another challenge is the complexity of correlation networks. When many assets are analyzed simultaneously, the resulting networks become dense and difficult to interpret, making it hard to distinguish significant relationships from less meaningful ones. In addition, many studies do not sufficiently evaluate the stability of clustering results across time, even though financial relationships evolve under different market conditions.

Therefore, there is a need for a structured probabilistic approach that improves robustness and interpretability. This project addresses these challenges by applying network filtering and probabilistic validation techniques, such as Minimum Spanning Trees and bootstrap reliability analysis, before hierarchical clustering is performed. The goal is to identify stable and economically meaningful groupings of financial assets that better represent true market structure.

4. Literature Review

Financial markets have long been recognized as complex systems in which asset prices move together through shared economic influences rather than independently. Early work by Mantegna (1999) demonstrated that correlations between asset returns can be transformed into distance measures that reveal a hierarchical organization of financial markets. This finding showed that assets naturally cluster according to economic similarity, providing a strong foundation for using hierarchical clustering to understand market structure. Later studies expanded this idea by showing that correlation structures evolve dynamically over time and that financial markets exhibit multiple structural layers, including market-wide, sector-level, and firm-level dependencies (Onnela et al., 2003; Tumminello et al., 2010).

As financial datasets became larger and more complex, researchers recognized that raw correlation matrices often contain significant noise, which can distort clustering results. To improve interpretability, network-based filtering approaches were introduced. Bonanno et al. (2003) proposed the Minimum Spanning Tree (MST) as a way to reduce network complexity while preserving the most important relationships among assets. Similar research showed that filtering techniques help isolate meaningful dependencies and create clearer representations of financial market structure (Kullmann et al., 2002; Pozzi et al., 2013).

In addition to filtering, probabilistic validation became a key focus of financial network research. Tumminello et al. (2007) demonstrated that many correlations observed in financial data are unstable across samples, emphasizing the importance of bootstrap reliability analysis. Bootstrap methods evaluate how consistently relationships appear across resampled datasets, allowing researchers to distinguish persistent connections from random fluctuations. This

probabilistic perspective strengthens confidence in clustering outcomes and improves methodological rigor in market analysis.

Network-based research has also contributed significantly to understanding systemic risk and diversification. Billio et al. (2012) showed that increasing interconnectedness among financial institutions contributes to systemic vulnerability, while Sandoval (2014) found that correlations tend to increase during periods of market stress, reducing diversification benefits. Additional studies demonstrated that network topology can influence how shocks spread across markets, further highlighting the importance of structural analysis in finance (Bardoscia et al., 2017).

More recent research confirms that graph-based and probabilistic approaches continue to play a major role in financial data science. Saha et al. (2022) reviewed modern network methods and emphasized their usefulness for identifying hidden market structure and improving analytical interpretation. Other studies using partial correlations and network spillover analysis further support the value of structured probabilistic models in financial research (Kenett et al., 2010; Diebold & Yılmaz, 2014).

Overall, the literature consistently shows that financial markets possess hierarchical structures that can be discovered through correlation-based network modeling, filtering, and clustering. However, many prior studies apply these techniques separately. This project builds upon existing research by integrating probabilistic filtering, bootstrap reliability testing, and hierarchical clustering into a unified framework designed to produce stable, interpretable, and economically meaningful market structures. Despite these advances, few studies integrate filtering, probabilistic validation, and hierarchical clustering into a unified framework, leaving a gap that this study addresses.

5. Project Purpose

The purpose of this project is to design and evaluate a structured probabilistic framework that improves the understanding of relationships among financial assets through hierarchical clustering. Financial markets are characterized by complex interdependencies in which assets often move together due to shared exposure to economic conditions, industry trends, and investor behavior. Traditional approaches that analyze assets independently frequently overlook these dependencies, making it difficult to interpret overall market structure or assess systemic risk.

This project specifically aims to identify latent hierarchical organization within financial markets by combining correlation analysis, probabilistic filtering, and clustering techniques. A secondary purpose is to improve the robustness of clustering outcomes through statistical validation so that results are less sensitive to noise and sampling variation. In addition, the project seeks to evaluate the temporal stability of relationships among assets, allowing analysis of how market structure changes across different market regimes. Ultimately, the study intends to provide insights relevant to diversification strategies and systemic risk analysis by identifying groups of assets that behave similarly under varying market conditions.

6. Research Question

The central research question guiding this study asks whether structured probabilistic models combined with hierarchical clustering can reveal stable and economically meaningful groupings among financial assets. This question arises from challenges commonly observed in financial clustering research, where traditional methods often produce unstable or difficult-to-interpret results due to noisy correlation estimates.

By introducing probabilistic filtering and validation techniques, this research investigates whether market structures identified through clustering become more consistent and aligned with economic intuition. Answering this question is important because reliable clustering results can provide deeper insight into portfolio diversification, risk concentration, and the transmission of shocks across financial systems.

7. Research Hypothesis

This study evaluates two primary hypotheses. The first hypothesis proposes that probabilistically filtered correlation networks will generate clusters that align with known economic sectors. Prior research suggests that companies operating within the same industry often respond similarly to macroeconomic factors, leading to stronger correlations among their returns. If probabilistic filtering successfully removes noisy relationships, the resulting clusters should display clear alignment with sector classifications.

The second hypothesis proposes that periods of increased market stress will lead to stronger correlations and tighter clustering structures. During volatile market conditions, assets often become more synchronized as investors respond collectively to uncertainty. This phenomenon reduces diversification effectiveness and should appear in the form of denser or more compact clusters within the hierarchical structure. Testing these hypotheses allows evaluation of both structural validity and practical financial implications of the proposed methodology.

8. Project Limitations (Expanded)

Several limitations should be acknowledged when interpreting the results of this study. First, correlation-based analysis measures statistical association but does not imply causal

relationships between assets. As a result, clustering outcomes should be interpreted as indicators of similarity rather than direct influence. Second, market relationships are dynamic and may change over time as economic conditions evolve, which means findings derived from one time period may not necessarily generalize to others.

Additionally, results may depend on asset selection and sample size. Different datasets or sector compositions may produce slightly different clustering structures. Finally, external shocks such as economic crises, policy changes, or unexpected global events may temporarily alter correlations and influence the stability of results. These limitations highlight the need for careful interpretation and motivate future extensions involving dynamic modeling.

9. Research Design

This project adopts a quantitative exploratory research design aimed at discovering structural patterns within financial markets. The study begins by collecting historical price data for selected financial assets. These prices are transformed into logarithmic returns to standardize measurements and allow consistent comparison across assets with different price levels.

Following data preparation, a correlation matrix is estimated to capture statistical relationships among assets. This matrix is then interpreted as a weighted network, allowing the market to be analyzed from a structural perspective. To reduce noise and improve interpretability, Minimum Spanning Tree filtering is applied, followed by bootstrap reliability analysis to validate edge stability. Correlations are converted into metric distances, enabling the application of hierarchical clustering algorithms. Finally, clustering outcomes are evaluated to examine economic interpretability and structural stability across time.

10. Participants and Sampling

In this study, participants refer to financial instruments rather than human subjects. The sample consists of selected stocks or exchange-traded funds representing multiple economic sectors, including technology, healthcare, finance, and industrials. This diversified sampling strategy ensures that clustering analysis captures both broad market behavior and sector-level relationships.

Sampling across sectors is particularly important because one of the research goals is to examine whether data-driven clusters align with known economic classifications. By including assets with diverse characteristics, the study improves the likelihood that identified structures reflect genuine market organization rather than narrow industry-specific trends.

11. Instruments for Data Collection

Data collection relies on historical price datasets obtained from reputable financial sources. These datasets provide daily closing prices used for return calculations. Several computational tools support the analysis process. Python libraries such as NumPy and Pandas are used for data handling, preprocessing, and transformation. Statistical computations, including correlation estimation, are performed using SciPy. NetworkX is used for constructing and analyzing financial networks, while Scikit-learn provides clustering algorithms and evaluation tools.

Together, these instruments enable efficient processing of financial data and support implementation of the probabilistic modeling framework.

12. Plan for Data Analysis

The data analysis process follows a structured sequence designed to support the research objectives. First, correlation analysis measures co-movement patterns among assets and forms

the basis for network construction. Since financial correlations are often noisy, Minimum Spanning Tree filtering is used to simplify the network while preserving essential relationships. This filtering step improves interpretability by removing redundant connections.

Bootstrap reliability analysis is applied to evaluate the statistical stability of network edges. A non-parametric bootstrap procedure is implemented by resampling daily observations (rows of the return matrix) with replacement while preserving the cross-sectional structure of assets. For each bootstrap sample, a new correlation matrix is computed and transformed into a distance matrix using the metric $dij = \sqrt{2(1 - \rho ij)}$. A Minimum Spanning Tree is then reconstructed based on the resampled distances.

This procedure is repeated B times (e.g., B = 500 or 1000). For each edge in the original MST, a stability score is calculated as the proportion of bootstrap networks in which the same edge appears. Edges with high occurrence frequency are interpreted as statistically robust relationships, while edges with low frequency are considered unstable and potentially driven by sampling noise. The distribution of edge stability scores provides a quantitative measure of network reliability. In addition, average edge stability and network persistence metrics (defined as the proportion of common edges between consecutive rolling windows) are computed to compare structural robustness across different market regimes.

Correlation values are transformed into distances to satisfy clustering requirements, and agglomerative hierarchical clustering is performed to reveal nested market structures. Finally, a rolling window approach (e.g., 252 trading days with a one-month step size) is implemented to evaluate temporal stability of network structure. This procedure allows examination of how

clusters evolve across different market conditions, providing insight into stability and structural change.

13. Plan for Presenting Results

Results will be presented using both visual and numerical approaches to support interpretation. Correlation heatmaps will provide an overview of co-movement intensity among assets. Network visualizations based on Minimum Spanning Trees will illustrate core relationships within the market structure. Hierarchical dendrograms will be used to display nested clustering relationships, enabling comparison across hierarchical levels.

Additional tables will summarize cluster composition and alignment with industry classifications. Stability metrics derived from rolling window analysis will be presented to demonstrate how clustering structures evolve over time. Together, these presentation methods aim to provide a clear and comprehensive understanding of the results.

14. Project Modifications

Several revisions were made following peer review and proposal modification assignments. One major improvement was clarifying the role of bootstrap reliability analysis as a central validation mechanism rather than a supplementary step. The methodological workflow was reorganized to improve clarity and logical progression, making it easier to understand how each step contributes to the research objective.

The project purpose was also rewritten to emphasize broader financial interpretation rather than technical terminology. In addition, stronger connections were made between clustering outcomes and practical implications such as diversification and systemic risk. These

modifications improved coherence and strengthened alignment between methodology and project goals.

15. Initial Results and Explanations

Preliminary findings suggest that clusters identified through the probabilistic framework generally align with known economic sectors, supporting prior research indicating hierarchical organization within financial markets. Minimum Spanning Tree filtering significantly reduces network complexity while maintaining key relationships, allowing clearer interpretation of market structure.

Bootstrap reliability analysis shows that core network connections remain stable across resampled datasets, suggesting that probabilistic filtering successfully reduces noise. Furthermore, initial analysis indicates that periods of higher market volatility are associated with stronger correlations and tighter clustering structures. This pattern suggests reduced diversification benefits during stressful market conditions, consistent with previous findings in financial network research. While these results are preliminary, they provide encouraging support for the proposed hypotheses.

16. Summary, Conclusion, and Future Research

This study demonstrates that combining structured probabilistic modeling with hierarchical clustering provides an effective and interpretable framework for analyzing financial market organization. By integrating network filtering and statistical validation, the methodology addresses key challenges related to noise and instability in financial correlation data. Initial findings suggest that the approach produces meaningful clustering structures aligned with economic interpretation.

Future research may extend this work by incorporating dynamic network models that adapt to changing market conditions, exploring alternative dependence measures such as partial correlations, and applying the framework across multiple international markets. Additional work may also examine how clustering results can be integrated into portfolio optimization and risk management strategies.

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